

Real-World Applications of Integers and Opposites

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Positive numbers are used to describe movement or situations in one direction, such as moving **up, forward, north, east**, earning money, or temperatures above zero.

Negative numbers are used to describe the opposite direction or situation, such as moving **down, backward, south, west**, losing money, or temperatures below zero.

A positive and a negative number with the same value represent **opposite quantities**.

Examples

Example 1

A submarine is 250 meters below sea level.

Below sea level is represented by a negative number.

$$\boxed{-250}$$

Example 2

A hiker climbs 180 meters above the starting point.

Above the starting point is represented by a positive number.

180

Example 3

The temperature is 9°C below zero.

Below zero is represented by a negative number.

$$\boxed{-9}$$

Example 4

A diver moves 15 meters downward from the water's surface.

Downward movement is represented by a negative number.

$$\boxed{-15}$$

Example 5

A person deposits ₱800 into a savings account.

A deposit is represented by a positive number.

800

Example 6

A student walks 12 meters west from the school gate. If east is positive, represent the movement using an integer.

West is the opposite direction of east.

$$\boxed{-12}$$

Example 7

An elevator goes up 7 floors from the ground floor.

Moving up is represented by a positive number.

7

Practice Exercises

Practice 1

Write a positive or negative number to represent a submarine that is 250 meters below sea level.

Practice 2

A hiker climbs 180 meters above the starting point. Represent the situation using an integer.

Practice 3

The temperature is 9°C below zero. Write the corresponding integer.

Practice 4

A diver moves 15 meters downward from the water's surface. Represent the movement with an integer.

Practice 5

A person deposits ₱800 into a savings account.
Represent the transaction using an integer.

Practice 6

A student walks 12 meters west from the school gate. If east is positive, represent the movement with an integer.

Practice 7

An elevator goes up 7 floors from the ground floor. Represent the movement using an integer.

Activities

Activity 1

Write a positive or negative number to represent a cave that is 180 meters below sea level.

Activity 2

A balloon rises 240 meters above the ground.
Represent the situation using an integer.

Activity 3

The temperature is 14°C below zero. Write the corresponding integer.

Activity 4

A diver descends 22 meters below the water's surface. Represent the movement using an integer.

Activity 5

A customer deposits ₦1 200 into a bank account. Represent the transaction using an integer.

Activity 6

A cyclist travels 20 meters west. If east is positive, represent the movement using an integer.

Activity 7

An elevator moves up 11 floors. Represent the movement using an integer.

Thank you!